

RUBIKSCUBESOLVER

User Manual

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About this manual. This guide walks you through installing and using the RUBIKSCUBESOLVER application that ships as a single-file executable on Windows and a .app bundle on macOS. The application implements the algorithms described in *Multi-Solution Search and the Limits of Neural Guidance in Rubik's Cube Solving*; this manual covers only the user-facing workflow, not the algorithmic details.

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1 What This Application Does

RUBIKSCUBESOLVER is an interactive desktop tool that accepts a 3-by-3 Rubik's Cube state from the user and returns a move sequence that solves it. Under the hood it runs an enhanced version of Kociemba's Two-Phase Algorithm with four selectable solver modes:

- **Baseline (K=1)** — the original Kociemba solver, fastest.
- **Multi-Solution (K=5)** — our primary contribution; explores five Phase 1 endpoints and keeps the shortest combined solution. On average cuts 0.53 moves off the baseline and roughly doubles the fraction of ≤ 20 -move solutions.
- **Anytime (K=5, 2s)** — the multi-solution solver with a 2-second soft budget, returning the best solution found within the budget. Designed for interactive applications where latency matters.
- **Neural LUT (K=5)** — multi-solution search augmented with a precomputed lookup-table move ordering. Included for completeness; quality matches plain K=5.

Input is either painted onto a 3D cube visualisation, loaded from one of four preset demo scrambles, or generated as a random scramble. Output is a human-readable move list that can also be played back one step at a time on the 3D view.

2 Getting the Application

2.1 Windows

Download `RubiksCubeSolver.exe` (single file, roughly 50 MB). Double-click to launch. No Python or other dependencies are required; everything is bundled inside the executable.

First launch. The first time you run the executable, Windows extracts bundled assets into a temporary directory. This takes a few seconds. Subsequent launches are faster.

SmartScreen warning. Because the executable is unsigned, Windows Defender SmartScreen may show a blue warning dialog on first launch. Click *More info* $\rightarrow$ *Run anyway*.

2.2 macOS

Download `RubiksCubeSolver.app.zip`, double-click to unzip, and drag the resulting `RubiksCubeSolver.app` to your Applications folder.

First launch (Gatekeeper). Because the app is unsigned, double-clicking it the first time will show “*cannot be opened because Apple cannot check it for malicious software*”. Work around this by *right-clicking* the app and choosing *Open*; macOS will then show an Open button you can click through. Subsequent launches work from a normal double-click.

3 The Main Window

When you launch RUBIKSCUBESOLVER you see four panels laid out in a single window:

- **3D view (top-left).** A rotatable 3D rendering of the cube. Left-drag to rotate, right-click a sticker to select it.
- **2D unfolded net (top-middle).** The standard cross layout showing all six faces at once. The currently selected sticker is highlighted.

- **Info panel (top-right).** Displays the selected sticker, the current solver mode, the control reference, and (after a solve) the move count and solution summary.
- **Colour palette (bottom-left).** Six swatches: White, Red, Blue, Yellow, Orange, Green. The currently selected colour is highlighted.

Four yellow *Demo* buttons and a *Random Scramble* button sit near the bottom-right. A single-line status message at the bottom of the window shows the most recent action or result.

4 Providing the Cube State

There are four ways to tell the solver what cube to solve. Pick whichever is most convenient.

4.1 Demo scrambles (recommended for first-time users)

Click any of the four yellow *Demo* buttons. Each loads a preselected valid cube state drawn from the paper’s seed-42 benchmark. The 3D view and unfolded net update immediately and the status bar reports “*Random scramble applied. Press Enter to solve!*” These are convenient for comparing the different solver modes on identical input: load the same demo twice, solve once with each mode, and compare the move counts.

4.2 Random scramble

Click *Random Scramble* (or press S) to apply 20–30 random moves to a solved cube. The result is a valid, uniformly distributed scramble. Use this when you want a fresh, unpredictable cube.

4.3 Painting stickers manually

If you want to solve a specific physical cube in your hand, you can paint the state one sticker at a time:

1. Orient your physical cube with **blue on top, red facing you, yellow on the right** (the orientation shown in the app’s header text).
2. Click a sticker on the 3D view or unfolded net to select it. The selected sticker is highlighted.
3. Pick a colour, either by clicking a swatch in the palette or by pressing a digit 1–6 (1:W, 2:R, 3:B, 4:Y, 5:O, 6:G). The selected sticker repaints instantly.
4. Press Tab / Shift+Tab to advance to the next / previous face, or use the arrow keys to move within a face.

Centre stickers are locked to prevent a common input error (they define which colour belongs to which face and must match the physical cube’s orientation).

4.4 Guided mode

Press G to toggle *guided mode*, which walks you through the cube sticker by sticker in a fixed order. Useful the first time you enter a cube by hand — each press of a number key paints the current sticker and advances automatically. Press G again to leave guided mode.

5 Choosing a Solver Mode

The info panel shows the currently active solver mode below the cursor position. Press **M** to cycle through the four modes:

Mode	Label	What it does
1	Baseline (K=1)	Kociemba’s original solver. Very fast (~ 10 ms on this build), slightly longer solutions.
2	Multi-Solution (K=5)	Our primary algorithm: tries 5 Phase 1 end-points, returns the shortest. Default mode.
3	Anytime (K=5, 2s)	Multi-solution search that always returns within 2 seconds using the best solution found so far.
4	Neural LUT (K=5)	Multi-solution search with a lookup-table move predictor. Quality matches K=5.

The **default mode is Multi-Solution (K=5)**. If you are just trying to solve a cube quickly and do not care about the algorithmic comparison, leave this alone.

6 Solving the Cube

Once a valid cube state is in place, press **Enter**. The solver runs on the current mode. Depending on the mode, a solve takes between ~ 10 ms (baseline) and ~ 15 s (K=5 on a hard cube). During a solve the window is unresponsive; wait for the status bar to update.

When the solve completes:

- The info panel shows the mode used, the move count, and the solution string.
- The status bar at the bottom reports success or failure.
- If no solution could be found within the internal time cap (15 s for K=5, 2 s for Anytime), you will see “*Error: Timeout, no solution within given time.*” Re-running the solve or switching to Baseline almost always works.

6.1 Stepping through the solution

Press **N** to advance one move. The 3D view animates the rotation and the status bar describes the move in plain English (for example “*U prime: top face counter-clockwise*”). Keep pressing **N** until the cube is solved. This is the core feature for using the app as a physical solver: make the move on your cube, then press **N** for the next one.

6.2 Comparing two modes on the same cube

A common demonstration workflow:

1. Click *Demo 1* to load a fixed scramble.
2. Press **M** until the info panel reads *Baseline (K=1)*.
3. Press **Enter**. Note the move count (e.g. 21).
4. Click *Demo 1* again to reset the same cube.

5. Press **M** until the info panel reads *Multi-Solution (K=5)*.

6. Press **Enter**. Note the move count (e.g. 18).

The 3-move saving on Demo 1 and Demo 2, or 2-move saving on Demo 3 and Demo 4, is the visible, reproducible demonstration of our contribution on a single cube. Across 200 random cubes the mean saving is 0.53 moves ($p < 10^{-20}$, paired test).

7 Keyboard and Mouse Reference

7.1 Solving and mode

Key	Action
Enter	Solve the current cube with the active mode.
N	Step to the next move in the solution.
M	Cycle through the four solver modes.
R	Reset the cube to solved.
S	Apply a random scramble (equivalent to the button).
Q	Quit the application.

7.2 Input and navigation

Key	Action
Tab	Move to the next face.
Shift+Tab	Move to the previous face.
↑ ↓ ← →	Move the cursor within the current face.
1 6	Paint the selected sticker with the corresponding colour.
G	Toggle guided-mode input.

7.3 Mouse

- **Left-drag on the 3D view:** rotate the view.
- **Click on a sticker (3D or 2D):** select that sticker.
- **Click on a colour swatch:** set the selected colour.
- **Click a Demo or Random Scramble button:** load a scramble.

8 Troubleshooting

“Windows protected your PC” dialog on first launch. The executable is unsigned. Click *More info* → *Run anyway*. Subsequent launches do not show this.

“Cannot be opened because Apple cannot check it” on macOS. Right-click the .app and choose *Open*. Then click *Open* in the follow-up dialog. This is a one-time click-through.

The app takes five seconds to show its window on Windows. Expected. PyInstaller single-file executables extract their contents to a temporary folder on first launch, which takes a few seconds. The window appears as soon as extraction finishes.

“Error: Timeout, no solution within given time.” The current cube is hard enough that the solver did not complete within the mode’s time cap. Options:

- Press **M** to switch to Baseline and press **Enter** again — Baseline uses the C backend and solves in milliseconds.
- Click the same Demo button again to reload the cube, then retry; solve time varies slightly run-to-run.

“Cannot change center piece!” Centre stickers are locked because they define which colour is assigned to which face. If the physical cube you are entering has a different centre colour arrangement than what the app shows, re-orient the physical cube to match (blue on top, red facing you, yellow on the right).

The 3D view and the unfolded net look out of sync. They are always derived from the same internal state. If they look different, the most likely explanation is that the 3D view has been rotated — only the camera changed, not the cube. Left-drag to rotate it back.

I pressed Enter and nothing happens. Check the status bar. If it shows an error about colour counts, the painted cube is invalid (there must be exactly nine stickers of each colour). Use **R** to reset and try again, or click a Demo button to load a known-valid state.

9 Where the Algorithm Comes From

The solver logic and the four modes correspond directly to sections of the accompanying paper, *Multi-Solution Search and the Limits of Neural Guidance in Rubik’s Cube Solving*. Readers who want the algorithmic detail behind a given mode should consult:

Mode in app	Paper section
Baseline (K=1)	Section 2.2 (Two-Phase Algorithm)
Multi-Solution (K=5)	Section 4.3 and Algorithm 1
Anytime (K=5, 2s)	Section 4.4
Neural LUT (K=5)	Section 4.5 and Section 6.8

This user manual does not duplicate that material. If you just want to solve a cube, everything you need is in Sections 1–7 above.